

The Magic Formula — Detailed Summary (Joel Greenblatt)

Source: Joel Greenblatt, *The Little Book That Still Beats the Market* (Wiley, 2010 updated edition) — read from the full text.

*Edition note: the book's own copyright page reads "Copyright © 2010 by Joel Greenblatt," ISBN 978-0-470-62415-9, and it contains an "Introduction to the 2010 Edition," an "Afterword to the 2010 Edition" (signed "—JOEL GREENBLATT, June 2010"), and results updated through 2009. Internally Greenblatt refers to the 2005 original as *The Little Book That Beats the Market* (which "ended up selling 300,000 copies worldwide and has now been translated into 16 different languages"). This deep summary is of the **2010 "Still" revision** and quotes it directly.*

Overview & thesis

The Magic Formula is a mechanical, two-factor value strategy: rank a whole universe of stocks on business **quality** (return on capital) and on **cheapness** (earnings yield), sum the two ordinal ranks, and buy the ~20-30 best-combined names. The one-sentence thesis, repeated verbatim throughout: the formula finds "**above-average companies** that we can buy at **below-average prices**" (Ch. 9). Greenblatt frames the whole book as a promise to his five children that he can "explain how to make money in terms that even my kids could understand" (Intro to the Original Edition), and insists the formula "will work for you only if you make the effort to fully understand *why* it works."

The intellectual lineage is explicit: Benjamin Graham supplies **Mr. Market** and the **margin of safety** (the "cheap" half); Warren Buffett and Charlie Munger supply the insight that "buying a business at a bargain price is great. However, buying a *good* business at a bargain price is even better" (Afterword). The Magic Formula operationalizes both into a single ranking a computer can run "in minutes every few months."

How the book is organized

Thirteen short chapters plus front and back matter. Chapters 1-3 build intuition: what a business is worth, the risk-free 6% government-bond hurdle, and reading an income statement (Jason's Gum Shops). Chapter 4 introduces **Mr. Market** and margin of safety. Chapter 5 defines the two tools — **earnings yield** and **return on capital**. Chapter 6 assembles the formula and reveals the headline back-test. Chapter 7 stress-tests it (size tiers, the decile study). Chapters 8-10 answer "why won't it get arbitrated away?" — the psychology of underperformance, capitalism/competitive advantage, and Mr. Market's long-run rationality. Chapters 11-12 cover DIY stock-picking and the "no tooth fairy" tour of Wall Street. Chapter 13 is a lighthearted wrap-up. The **Step-by-Step Instructions**, the **Afterword to the 2010 Edition** (updated numbers), and the **Appendix** (exact definitions and academic comparisons) close the book. Each chapter ends with a numbered "Quick Summary."

Core concepts (deep)

Mr. Market (Ch. 4). Greenblatt opens every semester by reading a stock's 52-week range aloud — e.g., "General Motors sold for \$37 per share yesterday. But over the last year, shares have sold for between \$30 and \$68." His point: "It makes no sense that the *values* of most companies swing wildly... during the course of each and every year," yet the *prices* clearly do. Why? "Who knows and who cares?... Maybe people just go nuts a lot (still my favorite)." Graham's device: you are partnered with a manic-depressive Mr. Market who each day quotes a price; you may buy, sell, or "very logically choose to do nothing." "You are never required to act."

Margin of safety (Ch. 4). Buying only at "a large discount to true value" — a \$37 price against a \$70 estimate — protects you if the estimate is wrong: "a purchase price of \$37 would leave enough margin for you to still make money" even if fair value were really \$50.

Earnings yield = cheapness (Ch. 5). Introduced as $\text{EPS} \div \text{price}$: Jason earns \$1.20/share at a \$12 price = a 10% earnings yield, compared against the risk-free 6% bond. "You would rather have a higher earnings yield than a lower one... Wax on!"

Return on capital = quality (Ch. 5). A Jason's store costs \$400,000 and earns \$200,000 = "a 50 percent return on capital." Contrast Jimbo's "Just Broccoli" store: \$400,000 cost, \$10,000 earnings = 2.5% — so bad "he shouldn't even be building" it. "You would rather own a business that earns a high return on capital than one that earns a low return on capital."

Why high ROC is "good" (Ch. 9). Two reasons. First, a firm that earns 50% can *reinvest* profits at 50% ("\$200,000 from the original store and an additional \$100,000 from the... new store" = 50% earnings growth). Second, sustained high returns imply "a special advantage of some kind" — new/better product, brand ("people will happily pay more for Coke than for Joe's Cola"), or competitive position (eBay's network) — "That special advantage keeps competitors from destroying the ability to earn above-average profits." Capitalism otherwise drives returns "down to average levels."

Why Mr. Market eventually pays fair value (Ch. 10). "Over the short term, Mr. Market acts like a wildly emotional guy... Over the long run... Mr. Market gets it right." Three mechanisms close the gap: smart investors buying bargains, companies buying back their own shares, and takeovers. Greenblatt "guarantee[s]" his MBA students that "two to three years is usually enough time for Mr. Market to get things right."

The formula & the exact definitions

The Appendix ("The Magic Formula") states the two measures precisely — and stresses why the textbook ratios are rejected.

1. Return on Capital = EBIT / (Net Working Capital + Net Fixed Assets). The denominator is "**tangible capital employed**." EBIT ("earnings before interest and taxes") replaces reported net income so companies "with different levels of debt and differing tax rates" compare cleanly. The denominator's construction: excess cash "not needed to conduct the business was excluded" from working capital; "short-term interest-bearing debt was excluded from current liabilities" (payables "are effectively an

interest-free loan"); net (depreciated) fixed assets are added. **Goodwill and other intangibles are excluded** — "Goodwill is a historical cost that does not have to be constantly replaced," so return on tangible capital "will be a more accurate reflection of a business's return on capital going forward." This is chosen over ROE and ROA, which are "distorted by ignoring the difference between *reported* equity and assets and *tangible* equity and assets."

2. Earnings Yield = EBIT / Enterprise Value, where EV = "market value of equity + net interest-bearing debt." Chosen over P/E or E/P because EV "takes into account both the price paid for an equity stake... as well as the debt financing." The worked example: Company A (no debt, priced \$60) and Company B (identical operations, \$50/share debt, priced \$10) have P/Es of 10 vs 3.33 and earnings yields of 10% vs 30% — yet **EBIT/EV is identical**: "You would be buying \$10 worth of EBIT for \$60, either way!" The office-building analogy makes the same point (\$1M building, \$800K mortgage: EBIT/EV = 10%, but pretax yield on the \$200K equity "would appear to be 26 percent").

3. Combined rank = rank(Return on Capital) + rank(Earnings Yield), ascending; take the top ~20-30. Greenblatt's own worked example (Ch. 6): a firm ranked "232nd best in return on capital and 153rd highest in earnings yield would receive a combined ranking of 385 (232 + 153)"; one ranked "1st in return on capital but only 1,150th... in earnings yield" gets 1,151. The formula deliberately never buys the top-ranked name on either axis alone — it wants "the best *combination* of those two factors."

Universe hygiene (Step-by-Step / Ch. 7). Market-cap floor (the largest-3,500 cutoff was "market value... of over \$50 million"; alternate tiers \$200M and \$1B; "for most individuals... above \$50 million or \$100 million"); **"Eliminate all utilities and financial stocks** (i.e., mutual funds, banks, and insurance companies)"; **"Eliminate all foreign companies...** the suffix 'ADR'." The general-screener proxy: "Use return on assets (ROA)... Set the minimum ROA at 25%" in place of ROC and screen "lowest price/earning (P/E) ratios" in place of earnings yield, and drop any stock with a P/E "of 5 or less" or that "announced earnings in the last week."

Back-test evidence (the numbers)

Headline, largest 3,500 (>\$50M), 1988-2004, 17 years, ~30-stock portfolio (Table 6.1): ~30.8% per year vs the market's **~12.3%**. "\$11,000 would have turned into well over \$1 million," against ~\$79,000 for the market.

Robustness to size (Ch. 7): largest 2,500 (>\$200M) returned **23.7%** vs **12.4%** ("practically doubled the market's average annual return"); largest 1,000 (>\$1B, buyable by "even the largest investors") again "practically double[d]" the market — the Appendix pegs that universe's market average at **11.7%**, with the best magic-formula decile at **18.88%** and the worst at **4.66%**.

Not luck: each of the three tests made "over 1,500 different stock picks"; combined, "over 4,500 separate magic formula selections." Performance was measured "over 193 separate one-year periods."

Monotonic decile test (2,500 names, held one year, Table 7.2, 1988-2004): Group 1 **17.9%**, G2 15.6, G3 14.8, G4 14.2, G5 14.1, G6 12.7, G7 11.3, G8 10.1, G9 5.2, G10 **2.5%** — "Group 1 beats Group 2, Group 2 beats Group 3... straight down the line," Group 1 beating Group 10 "by over 15 percent a year." Because it "is merely a ranking formula... there always have to be stocks that rank at the top" (unlike Graham's net-net formula, which "few, if any, companies qualify for" today).

Consistency (Ch. 10): over any three-year period (largest 1,000), the formula "beat the market averages **95 percent of the time** (160 out of 169... periods)" and made money **100% of the time** (169 of 169). The "worst return... was a gain of 11 percent," versus the market's worst three-year "loss of 46 percent." For the largest 3,500, "every three-year period... beat the market averages (169 out of 169)."

Updated through 2009 (Afterword): over 1999-2009 — "one of those rare 10-year periods over which the S&P 500 index was actually down" — the largest-1,000 test "earn[ed] 255 percent... a 13.5 percent annualized return during a 10-year period when the S&P index was actually down 0.9 percent per year." The refreshed decile (Table A.2, 1988-2009): G1 **15.2%**, G2 12.7, G3 12.1, G4 11.5, G5 10.7, G6 10.2, G7 8.8, G8 7.1, G9 4.1, G10 (**0.2%**). Over these 22 years the portfolios "captured" **95% of the S&P's down-month performance and 140% of its up-month performance.**

Methodology defense (Appendix, "A Random Walk Spoiled"): the study used S&P Compustat's "Point in Time" database, eliminating look-ahead and survivorship bias; the two factors "were actually the first two factors tested" (no data mining). A pure size effect is ruled out — by market-cap decile, the smallest 10% returned 12.1% and the largest 11.9%. Against price/book (large caps), the cheap-minus-expensive decile spread was "only 2 percent per year," versus the Magic Formula's ~14%. Piotroski's 9-point F-score and the Haugen-Baker 71-factor model are shown to fade for large caps where the Magic Formula "remain[s] incredibly robust."

Chapter / thematic walkthrough

- **Ch. 1-3 (intuition):** the gum-selling superhero Jason teaches present value, the 6% "no risk" hurdle, and the income statement (Jason's Gum Shops: \$10M sales → \$1.2M net income; \$12/share → 10% earnings yield).
- **Ch. 4 (Mr. Market):** prices ≠ values; margin of safety; three lingering doubts (how to value a business; how to know you're right; how to beat the pros).
- **Ch. 5 (the two tools):** *The Karate Kid* "wax on/wax off" — earnings yield (higher is better) and return on capital (Jason 50% vs Jimbo/Just Broccoli 2.5%). "Don't give money to guys named Jimbo."
- **Ch. 6 (the formula):** rank 3,500 names on each factor, sum, buy the top ~30; the 30.8% result. "It appears to work too well."
- **Ch. 7 (stress test):** size tiers and the monotonic decile — "the magic formula appears to work *in order*."
- **Ch. 8 (why it survives):** it "doesn't work at all" often — underperforms in "5 out of every 12 months," fails once every four years, and "for one out of every six periods... did poorly for more than two years in a row." The best-selling-author-turned-fund-manager and the "smartest money manager I know" (down to "only four original clients") illustrate why most quit — which is *why* the edge persists.
- **Ch. 9 (timeless logic):** high ROC signals reinvestment opportunity and competitive advantage; the formula finds "above-average companies at below-average prices."
- **Ch. 10 (Mr. Market gets it right):** over three years the formula "never lost money"; risk redefined as (1) chance of long-term loss and (2) chance of trailing alternatives.
- **Ch. 11 (DIY):** last year's earnings are a crude proxy; ideally use "normal" (normalized) earnings 3-4 years out. "Most people have no business investing in individual stocks on their own!" If you must,

concentrate in "five to eight" well-understood names, or pick "from the top 50 or 100" ranked stocks (10-30 holdings).

- **Ch. 12 ("no tooth fairy"):** brokers sell, most mutual funds trail after fees ("average returns minus fees equals below-average returns"), past records don't predict, hedge funds charge "1 percent... plus a 20 percent share of the profits"; index funds settle for the market's ~10%/yr. The only real edge left "rhymes with... basic formula."
- **Ch. 13 & Step-by-Step:** the mechanics; the Afterword adds the two "serious flaws" — it "can underperform the market for years," and "**Beating the market is not the same thing as making money**" (100% long, so it falls with the market).

How it maps to the Trader Screener

1. **§2d "Composite score (model.json)" is the direct analog of the entire rank-and-combine engine.** The Magic Formula *is* a two-factor composite; it is the cleanest template for how model.json should fold factors into one sortable number — via **summed ordinal ranks** ($\text{rank}(A) + \text{rank}(B)$), which sidesteps the unit mismatch between a percentage and a yield. Worked example to mimic: $232 + 153 = 385$.
2. **§2d "ROIC" $\approx$ Return on Capital.** The screener's quality axis. Fidelity caveat: Greenblatt's is strictly $\text{EBIT} / (\text{Net Working Capital} + \text{Net Fixed Assets})$ with excess cash, short-term debt, and goodwill removed — stricter than a generic book-value ROIC field.
3. **§2d "P/E" $\approx$ Earnings Yield — but upgrade it.** Greenblatt explicitly rejects P/E for $\text{EBIT} / \text{Enterprise Value}$ (his Company A/B example), so the screener should carry an **EBIT/EV** column beside §2d "P/E" as the capital-structure-neutral cheapness rank.
4. **§2d "Market cap / cap size" is exactly the cap-floor filter** (\$50M / \$200M / \$1B; \$50M-\$100M for individuals) — used as a **pre-rank filter**, not a display column.
5. **§2d "Sector" implements the exclusions** — drop financials (banks, insurers, mutual funds) and utilities, "whose balance sheets break the capital/EBIT math," plus foreign/ADR names, *before* ranking.
6. **§2c "Dollar volume (ADV $\times$ price)" / §2c "Average daily volume (ADV)"** make explicit the liquidity/tradeability screen the formula otherwise handles only implicitly through its cap floor — Greenblatt's whole rationale for a size floor is that thin names "can't be replicated" in the real world.

Contrast with the pipeline's DCF/Comps/EPV model. §2d "Upside % (DCF/comps)" is an *absolute* intrinsic-value estimate requiring forecasts, a discount rate, and WACC — precisely the "normalized earnings several years out" work Greenblatt says most people "have no business" doing. The Magic Formula is purely *relative and cross-sectional*: current (trailing) financials only, no projections, ranked against peers. They are complementary — EPV/DCF answers "what is it worth?"; the Magic Formula answers "which is best-ranked right now?" — and §2d "Composite score (model.json)" can host both as separate factors.

Actionable takeaways

- Build §2d "Composite score" as **summed ordinal ranks**, seeded with §2d "ROIC" (ideally EBIT/tangible-capital) plus a new **EBIT/EV** earnings-yield column, before adding momentum (§2a) or smart-money (§2e) factors.
- Apply §2d "Market cap / cap size" and §2d "Sector" (financials/utilities/ADR) as **pre-rank filters**; add a §2c dollar-volume gate as an explicit liquidity floor.
- Treat the output as a **portfolio-of-many** signal — 20-30 equal-weight names, one-year rolling holds (Greenblatt phases in "five to seven... every two to three months"), and a **3-5 year minimum commitment** — never a single-stock trigger.
- Surface an "underperformance is normal" note in the UI: the strategy trails ~1 year in 4 and can lag "for years," which is the very reason it keeps working.

Caveats, limits & what to distrust

- **It is 100% long.** "Beating the market is not the same thing as making money" — a 38% loss beating a 40% market drop is "small consolation" (Afterword).
- **Long stretches of underperformance are structural, not incidental** — 5 of 12 months, 1 in 4 years, some 3-year lags; even the great 1999-2009 window contained ~4 years (34 + 13 months) of trailing the S&P. Only 50-60% of top-ranked names actually beat the market; the edge is an *average*.
- **Trailing earnings are a blunt instrument** — great across a 20-30 name portfolio, unreliable for any single stock; do not read one Magic Formula pick as a valuation.
- **No shorting the bottom decile.** The obvious long-G1/short-G10 trade "would have gone broke — losing 100 percent of our money" around 2000; the formula "doesn't work and play well with others."
- **The website is not the method.** magicformulainvesting.com merely reproduces the study with clean data; "the mechanics aren't the important part." Note the behavioral tell: given a choice, "97 percent" chose to have it implemented for them.
- **Definition fidelity matters.** Loose ROIC/P/E fields are only a *rough approximation* of Greenblatt's EBIT-based ratios.

Open questions

- Does the pipeline's §2d "ROIC" use EBIT/tangible-capital (excess cash, short-term debt, goodwill removed), or a looser book definition? Fidelity to the back-test depends on it.
- How should §2e smart-money and §2a momentum enter the same rank-sum without one factor swamping the two Greenblatt factors — equal-weight ranks or weighted?
- What rebalance cadence fits a nightly free-data pipeline versus Greenblatt's annual, tax-timed, rolling five-to-seven-per-quarter cycle?
- Exclude financials/utilities globally, or only for the factors their accounting distorts (ROC/EBIT), while keeping them eligible on other axes?

- Trailing vs "normalized" EBIT: worth the added complexity in a free pipeline, or does a many-name portfolio wash out the noise as Greenblatt argues?

Sources

- **Primary:** Joel Greenblatt, *The Little Book That Still Beats the Market*, updated edition (John Wiley & Sons, 2010; ISBN 978-0-470-62415-9) — summarized directly from the full book text.
- **Sections referenced:** Introduction to the 2010 Edition and Introduction to the Original Edition (goals, "must make sense to *you*"); Ch. 1-3 (Jason's Gum Shops, the 6% hurdle, the income statement); Ch. 4 (Mr. Market, margin of safety); Ch. 5 (earnings yield and return on capital); Ch. 6 & Table 6.1 (the formula; largest-3,500, 30.8% vs 12.3%); Ch. 7 & Tables 7.1-7.2 (the \$200M/\$1B tiers, 23.7% vs 12.4%, and the monotonic decile); Ch. 8-10 (underperformance psychology, competitive advantage, Mr. Market's long-run rationality, the 95%/100% three-year statistics); Ch. 11-12 (DIY limits, "no tooth fairy"); the **Step-by-Step Instructions** (20-30 holdings, one-year holds, tax timing, financials/utilities/ADR exclusions, ROA-25% general screen); the **Afterword to the 2010 Edition** & Tables A.1-A.2 (255%/13.5% over 1999-2009, the 1988-2009 decile, up/down capture, the two "flaws"); and the **Appendix ("The Magic Formula" / "A Random Walk Spoiled")** for the exact EBIT-based definitions and the Point-in-Time methodology.